

MUHAMMAD FACHRIL FIRMANSYAH

Software Engineer

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PROFILE

Final-Year Computer Science student specializing in Artificial Intelligence with hands-on experience in Software Engineering, Machine Learning, Computer Vision, and Full-Stack Development. Skilled in building AI-powered applications and web systems using Python, JavaScript, TypeScript, React.js, Next.js, Flask, and modern machine learning frameworks. Experienced in developing projects involving object detection, image classification, data processing, and decision support systems. Passionate about solving real-world problems through practical technology solutions and continuously developing technical skills.

EDUCATION

Binus University **2023 - Present**

Undergraduate of Computer Science | GPA : 3.85

Streaming : Artificial Intelligence

EXPERIENCE

Software Engineer Intern

PT Indofood CBP Sukses Makmur Tbk - Packaging Division | Tangerang **(August 2026 - Present)**

- Developed full-stack web applications using React.js, Node.js, Express.js, PostgreSQL, and Tailwind CSS for inventory and accounting monitoring.
- Built inventory and invoice management modules with CRUD operations, search, filtering, and data validation for cylinder stock and accounting invoices.
- Designed responsive dashboards, reusable UI components, forms, and data tables, and integrated RESTful APIs with PostgreSQL.
- Performed debugging, functional testing, error handling, and performance optimization to ensure application reliability and data consistency.

Web Developer & Mathematics Teacher

SMK Penerbangan Dirghantara | Tangerang **(January 2025 – July 2026)**

- Spearheaded the end-to-end development, deployment, and maintenance of the school's official profile website.
- Built responsive web pages and optimized front-end layout and performance to ensure information is delivered effectively to users.
- Fostered analytical and logical problem-solving skills in students through mathematics instruction.

Informatics & Artificial Intelligence Teacher

SMA Pramita | Tangerang **(July 2025 – July 2026)**

- Taught Informatics and Artificial Intelligence, covering programming, algorithms, data cleaning, machine learning, and AI fundamentals.
- Guided students through data preprocessing, model development, coding, debugging, and problem-solving using practical projects.
- Designed hands-on learning activities to develop students' programming, data analysis, and computational thinking skills.

Web Developer

Pemoeda.ofc | Jakarta **(June 2024 – 2025)**

- Engineered highly scalable and responsive web applications using modern JavaScript frameworks, focusing on delivering an intuitive and seamless user experience.
- Optimized frontend performance, page load speed, and cross-device compatibility to ensure high-quality web interfaces.
- Wrote clean, modular, and maintainable code while focusing on UI/UX improvements and usability.

PROJECTS

AsphaltEye

AI-Powered Road Damage Detection & Community Reporting Platform **(June 2026 – July 2026)**

- Developed an AI-powered road damage detection platform using YOLOv11, Python, Flask, and React.js for automated pothole and crack detection
- Prepared datasets and trained YOLOv11 models for multi-class road damage classification and defect counting
- Built and integrated a Flask REST API for AI inference, image processing, and real-time detection with the React.js frontend
- Developed a GPS-tagged community reporting system with photo-based road damage reports and location visualization
- Optimized AI inference and API processing for lower latency and improved application performance during real-time media analysis

Tanara

SIBI Sign Language Detector **(April 2026 – June 2026)**

- Developed a real-time SIBI sign language recognition system using MobileNetV2, TensorFlow/Keras, MediaPipe, OpenCV, and Flask
- Trained a MobileNetV2 classification model to recognize 36 SIBI gestures (A–Y and 0–9) from hand landmark and image data
- Integrated MediaPipe Hands for real-time hand tracking and preprocessing before AI inference
- Built a Flask REST API for real-time model inference and connected it with a React.js + Vite frontend
- Implemented webcam and media-upload detection, text-to-sign visualization, and a gamified 60-second SIBI quiz
- Optimized the lightweight AI inference pipeline to improve real-time responsiveness and prediction performance under varying lighting and background conditions

HemaVision

Automated Blood Cell Analyzer **(February 2026 – April 2026)**

- Engineered an AI-powered blood cell analysis system using YOLOv11, PyTorch, OpenCV, Flask, and React.js to detect and count 3 blood cell types: RBCs, WBCs, and platelets
- Trained and optimized YOLOv11 for automated detection from microscopic blood smear images, replacing manual cell-counting workflows
- Developed Flask REST APIs supporting both single-image and batch inference, enabling scalable blood sample processing
- Implemented interactive cell-count visualizations and automated PDF medical report generation, streamlining analysis and reporting workflows

Sistemku

School Attendance & Decision Support System **(January 2026 – March 2026)**

- Engineered a cloud-ready school attendance and Decision Support System (DSS) using Next.js, TypeScript, Drizzle ORM, Turso, and Tailwind CSS.
- Digitized paper-based attendance and integrated 3 evaluation criteria: attendance, academic performance, and discipline for objective student assessment.
- Implemented the Simple Additive Weighting (SAW) algorithm to automatically normalize, weight, and rank student performance.
- Developed real-time dashboards with Excel/PDF export capabilities, streamlining attendance tracking, student ranking, and administrative reporting.
- Delivered the system using an Agile-Scrum development approach, managing the full development lifecycle from requirements analysis to deployment.

Rekita

Smart Waste Management App **(February 2025 – March 2025)**

- Designed an AI-powered mobile waste management application using Figma, FigJam, Adobe Illustrator, and User Research, integrating AI image recognition, GPS mapping, and gamification.
- Conducted user research and applied the Design Thinking methodology through Empathize, Define, Ideate, Prototype, and Test phases.
- Developed user personas, Affinity Diagrams, Prioritization Matrices, HMW questions, and Crazy 8s to define user needs and validate design decisions.
- Prototyped end-to-end user flows for AI waste scanning, GPS-based recycling center discovery, and Eco-Points gamification.
- Delivered a high-fidelity interactive prototype with an intuitive information architecture and streamlined user journey from waste identification to recycling drop-off.

TECHNICAL SKILLS AND INTERESTS

- **Languages:** Python, JavaScript, TypeScript, C++, SQL
- **Frameworks & Libraries:** React.js, Next.js, Node.js, Express.js, Flask, TensorFlow, PyTorch, OpenCV, MediaPipe
- **Databases:** MySQL, PostgreSQL, SQLite, Turso
- **Tools:** Git, GitHub, VS Code, Figma, FigJam, Vite
- **AI & Machine Learning:** Computer Vision, Object Detection, Image Classification, Machine Learning, Deep Learning, YOLOv11
- **Areas of Interest:** Software Engineering, AI Engineering, Machine Learning, Computer Vision, Full-Stack Development
- **Soft Skills:** Problem Solving, Analytical Thinking, Communication, Adaptability, Self-Learning, Teamwork